Project Milestone 02

Group Name: 2by2	Lab Section: EF - 6L
Member 1: Abucay, Mark Leo	Member 2: Labramonte, Nathalie Faith
Member 3: Reyes, Thomas III	Member 4: Solmoro, Kriztian Mer

Data Preprocessing:

- filereader.py:
    - main file used for data-loading and preprocessing for all four methods (Naive Bayes Classifier, K-Nearest Neighbors (KNN), Perceptron, and K-Means Clustering)
- Operations Performed:
    - takes a zip file containing the necessary CSV files for the data in comparison as an input
    - removes leading and trailing whitespaces from column headers and converts them into lowercase
    - uses utf-8-sig encoding in order to handle UTF-8 files that may contain Byte Order Mark (BOM)
    - skips blank rows or invalid entries, nonnumeric values, and non-finite values such as NaN and infinity
    - identifies the region and months columns
    - gets the sum of locality records based on region, month and year
- Naive Bayes, KNN, Perceptron:
    - each of the observations represents the monthly tax collection of a specific region within a given year
    - Target Variable: Binary Classification
    - 2 Classes of the Target Variable:
        - Up: increase of tax collection compared to the same month of the previous year
        - Down: decrease or no change in the tax collection compared to the same month of the previous year
    - contains 21 input features:
        - 18 Regions: one-hot encoded features wherein each observation is represented using binary values
        - 2 Months: represented as sine and cosine transformations of the month in order to showcase its cyclical nature
        - 1 Historical Collection: the amount of tax collection for the same region and month in the past year
- K-Means Clustering:
    - aggregated into one observation per region with the use of collection history from 2020 up to 2024

- represented using 3 features:
  - Mean Monthly Collection: average monthly tax collection within the five-year period
  - Average Annual Growth Rate: average of the year-over-year percentage change in total annual collections from 2021 up until 2024
  - Collection Volatility: standard deviation of monthly collections over absolute mean monthly collection

Train/Test Split:

- Training Data:
  - 2021 - 2023 collections
  - Total: 648 records (18 regions * 12 months * 3 years)
- Testing Data:
  - 2024 collections:
  - Total: 216 records (18 regions * 12 months)
- Time-Based Split:
  - used for Naive Bayes, KNN, Perceptron
  - to be able to predict whether collections rise (go up) compared with the previous year, mimicking actual forecasting

Evaluation Metrics:

- Naive Bayes, KNN, Perceptron:
  - Accuracy:
    - $\frac{TP + TN}{TP + TN + FP + FN}$
    - the fraction of all predictions that are correct in each model
    - provides an overall measure of the performance
  - Precision:
    - $\frac{TP}{TP + FP}$
    - examines how many of the predicted “up” cases were actually “up”
    - higher precision insinuates that there are fewer false positive predictions
  - Recall:
    - $\frac{TP}{TP + FN}$
    - examines how many of the actual “up” data were successfully identified
    - higher recall indicates that there are fewer missed increases in tax collection
  - Macro-F1 Score:
    - $\frac{F1_{Up} + F1_{Down}}{2}$ where F1 = $2\ x\ \frac{Precision\ x\ Recall}{Precision + Recall}$

        - gives equal importance to “up” and “down” cases even when they have different number of observations
        - can balance out the number of ups and downs
    - Confusion Matrix:
        - a 2 x 2 matrix is used in order to track examine the types of correct and incorrect predictions

Actual/Predicted	Down	Up
Down	True Negative (TN)	False Positive (FP)
Up	False Negative (FN)	True Positive (TP)

        - identifies the type of mistakes made by each model
- K-Means Clustering:
    - Silhouette Score:
        - For each point, s = (b-a) / max(a,b), where a is the average distance to its own cluster and b is the average distance to the nearest other cluster. It ranges from -1 to 1, and higher values means clusters are separated better. Report the average over all points, and compare it across values of k.
    - Inertia (WCSS) with Elbow Method:
        - Inertia measures the sum of squared distances from data points to their assigned centroids, continually decreasing as the number of clusters increases (meaning tighter groupings). To determine the optimal number of clusters rather than score overall model quality, the elbow method plots this inertia against cluster count so you can select the exact point where the curve bends and diminishing returns flatten out.

Initial Results:

- Naive Bayes Classifier

```
Enter k: 1

k = 1.0


-----------CONFUSION MATRIX-----------
                Predicted
                Down   Up
Actual Down        0   34
Actual Up         12  170

----------EVALUATION METRICS----------
Accuracy:        0.7870
Macro Precision: 0.4167
Macro Recall:    0.4670
Macro F1-Score:  0.4404

-----------PER-CLASS METRICS-----------

Class: Down
Precision: 0.0000
Recall:    0.0000
F1-Score:  0.0000

Class: Up
Precision: 0.8333
Recall:    0.9341
F1-Score:  0.8808
```

- K-Nearest Neighbors (KNN)

```
Classifying test instances...

KNN CLASSIFICATION RESULTS
k = 3
p = 2.0


----------CONFUSION MATRIX----------
                Predicted
                Down   Up
Actual Down        0   34
Actual Up         10  172

----------EVALUATION METRICS----------
Accuracy:        0.7963
Macro Precision: 0.4175
Macro Recall:    0.4725
Macro F1-Score:  0.4433

----------PER-CLASS METRICS----------

Class: Down
Precision: 0.0000
Recall:    0.0000
F1-Score:  0.0000

Class: Up
Precision: 0.8350
Recall:    0.9451
F1-Score:  0.8866
```

- Perceptron

```
Final Adjusted Weight: 0.3000, 0.0000, 0.0000, -0.2000, 1.6000, -0.6000, -0.4000, 0.0000, 0.0000, 0.0000, 0.0000, -0.2000, 0.0000, 0.0000, 0.0000, 0.0000, -0.2000, 0.0000, -0.2000, 0.0194, -0.01
82, -3.4841
Training did not converge within 100 epochs
Results saved as csv file

----------CONFUSION MATRIX----------
                Predicted
                Down   Up
Actual Down        4   30
Actual Up         28  154

----------EVALUATION METRICS----------
Accuracy:        0.7315
Macro Precision: 0.4810
Macro Recall:    0.4819
Macro F1-Score:  0.4814

----------PER-CLASS METRICS----------

Class: Down
Precision: 0.1250
Recall:    0.1176
F1-Score:  0.1212

Class: Up
Precision: 0.8370
Recall:    0.8462
F1-Score:  0.8415
```

- K-Means Clustering

```
Number of Regions: 18
Number of Features: 3

Enter the number of clusters, k (0 to pick the best k from 2 to 10): 2

Total # of regions: 18
Features: log10 of mean monthly collection, average annual growth, volatility (standardized)

################################################

K SELECTION
k    Inertia       Silhouette
--------------------------------
2    29.30         0.4339
3    15.32         0.4619
4    9.46          0.4119
5    7.04          0.3965
6    5.32          0.3772
7    4.24          0.3135
8    2.60          0.3541
9    2.17          0.3223
10   1.34          0.3063

################################################

Finding k=2 clusters...

Number of Clusters: 2
Inertia: 29.30
Inertia over 10 restarts: min 29.30, max 44.02
Converged in 3 iterations


----------CLUSTER PROFILES----------
Cluster  # Regions  Mean Monthly (M PHP)  Avg Growth (%)  Volatility
0        12         15,050.25             15.24           0.314
1        6          1,548.19              21.77           0.361

----------REGIONS PER CLUSTER----------
Cluster 0: Bangsamoro Autonomous Region in Muslim Mindanao (BARMM), MIMAROPA Region, National Capital Region (NCR), Negros Island Region (NIR), Region I (Ilocos Region), Region II (Cagayan Valley), Region
IV-A (CALABARZON), Region IX (Zamboanga Peninsula), Region VI (Western Visayas), Region VII (Central Visayas), Region X (Northern Mindanao), Region XI (Davao Region)
Cluster 1: Cordillera Administrative Region (CAR), Region III (Central Luzon), Region V (Bicol Region), Region VIII (Eastern Visayas), Region XII (SOCCSKSARGEN), Region XIII (Caraga)

Overall Silhouette score: 0.43387872752406415
```